Generative AI Interview Questions and Answers

Simple answers you can say in your own words
Generative AI / AI Engineer interviews

How to use this: Read each question, answer out loud first, then compare with the answer here. The 'Say in interview' box is a short version you can speak in 20 to 30 seconds. These are practice questions based on common topics, not leaked questions from any company.

Part 1: Basics

Q1. What is Generative AI?

Answer:

Generative AI is AI that creates new content, like text, images, code, audio, or video. It learns the patterns from a large amount of data, and then it makes new content that looks similar. Examples: ChatGPT, Claude, and Gemini for text, and Stable Diffusion or DALL-E for images.

Say in interview: Generative AI creates new content like text, images, and code by learning patterns from large data.

Q2. What is the difference between generative AI and traditional AI?

Answer:

Traditional (discriminative) AI looks at the data and decides something, like 'is this email spam or not' or 'is this a cat or a dog'. Generative AI creates something new, like writing an email or drawing a cat.

- Traditional AI: classify, predict, detect.
- Generative AI: write, draw, summarize, answer, create.

Say in interview: Traditional AI answers 'what is this', and generative AI answers 'create something like this'.

Q3. What is an LLM?

Answer:

LLM means Large Language Model. It is a very big neural network trained on a huge amount of text. It understands and writes human language. 'Large' means it has billions of parameters (the numbers learned during training). Examples: GPT, Claude, Gemini, and Llama.

Say in interview: An LLM is a very large model trained on huge text data that can understand and generate language.

Q4. How does an LLM generate text?

Answer:

An LLM predicts the next token again and again. It reads the text so far, calculates the probability of every possible next token, picks one, adds it to the text, and repeats. This continues until the answer is complete. So it writes one token at a time.

Say in interview: An LLM predicts the next token one at a time, based on all the text before it, and repeats until the answer is done.

Q5. What is a token?

Answer:

A token is a small piece of text, which can be a whole word, part of a word, or a symbol. The model does not read letters or words, it reads tokens. As a rough idea, 100 tokens are around 75 English words. We pay for the API by tokens, and the model limits are also in tokens.

Say in interview: A token is a piece of text the model reads and writes. Cost and limits are counted in tokens.

Q6. What is a context window?

Answer:

It is the maximum number of tokens the model can handle at one time. It includes our prompt, the documents, the chat history, and the answer together. If the text is longer than that, we must cut it or summarize it. A bigger context window means the model can read more, but it costs more and can be slower.

Say in interview: The context window is the most tokens the model can see at one time, input and output together.

Q7. What are open-source models and closed (proprietary) models?

Answer:

Closed models, like GPT, Claude, and Gemini, are used through an API. We do not get the model itself. Open-weight models, like Llama and Mistral, can be downloaded and run on our own servers.

- **Closed:** easy to start, usually top quality, but data goes to the provider and we pay per use.
- **Open:** more control and privacy, we can fine-tune them, but we must manage the servers and the cost.

Say in interview: Closed models are easy to use through an API. Open models give more control and privacy but need our own setup.

Part 2: Transformers

Q8. What is a transformer?

Answer:

The transformer is the neural network design behind almost all modern LLMs. It was introduced in the 2017 paper 'Attention Is All You Need'. Its main idea is the attention mechanism, which lets the model look at all the words in the text at the same time and decide which words are important for each other.

Say in interview: A transformer is the model design behind modern LLMs. It uses attention to understand how all the words in a text relate to each other.

Q9. What is the attention mechanism?

Answer:

Attention lets the model decide which other words to focus on when it processes a word.

Example: In the sentence 'The dog did not eat the food because it was tired', the word 'it' should focus on 'dog' and not on 'food'. Attention gives a higher score to 'dog' for the word 'it'.

Say in interview: Attention lets each word look at all the other words and focus on the ones that matter most for its meaning.

Q10. Why are transformers better than RNNs?

Answer:

RNNs read the text one word after another, so they are slow to train, and they forget the early words in long text. Transformers have these advantages:

- They process all the words in parallel, so training is much faster on GPUs.
- They handle long-range relations better, because every word can directly look at every other word.
- They scale well to very big models and data.

Say in interview: Transformers work in parallel and handle long text better, while RNNs are slow and forget the early words.

Q11. What is positional encoding?

Answer:

Attention alone does not know the order of the words. 'Dog bites man' and 'Man bites dog' would look the same. So we add information about the position of each token to its embedding. This way the model knows the word order.

Say in interview: Positional encoding adds the position information, so the model knows the order of the words.

Q12. What is the difference between encoder and decoder models?

Answer:

Transformers have an encoder part and a decoder part.

- **Encoder-only (like BERT):** reads the whole text and understands it. Good for classification, search, and embeddings.
- **Decoder-only (like GPT, Claude, Llama):** generates text one token at a time. Most modern LLMs are this type.
- **Encoder-decoder (like T5):** reads an input and generates an output. Good for translation and summarization.

Say in interview: Encoder models understand text, decoder models generate text, and most LLMs today are decoder-only.

Q13. What are embeddings?

Answer:

An embedding is a list of numbers that represents the meaning of a text. Similar meanings have similar numbers. They are used in search, RAG, recommendation, and clustering.

Say in interview: Embeddings are numbers that represent meaning, so we can compare texts by similarity.

Part 3: Training and Customizing

Q14. How is an LLM trained? (pre-training, fine-tuning, RLHF)

Answer:

There are usually three stages:

- **Pre-training:** the model reads a huge amount of text from the internet and books and learns to predict the next token. It learns language and general knowledge. This is very costly.
- **Instruction fine-tuning:** we train it on examples of questions and good answers, so it follows instructions and behaves like an assistant.
- **Alignment (RLHF or similar):** humans rate the answers, and the model learns to give answers that are helpful, honest, and safe.

Say in interview: First it learns language by predicting the next token on huge data. Then it is trained to follow instructions. Then it is aligned with human feedback to be helpful and safe.

Q15. What is fine-tuning? When do we need it?

Answer:

Fine-tuning means training an already trained model more on our own smaller data, so it works better for our specific task. We need it for a fixed style or format, a special domain language, or to make a small model do one task well. It is not the best way to add new facts. For facts we use RAG.

Say in interview: Fine-tuning trains a ready model on our data for a specific style or task. For new facts I prefer RAG.

Q16. What is LoRA and PEFT?

Answer:

Full fine-tuning changes all the billions of parameters, which needs a lot of GPU and memory. PEFT (Parameter Efficient Fine-Tuning) trains only a very small part. LoRA is the most popular PEFT method. It keeps the original model frozen and adds small extra layers, and only those are trained. It is much cheaper and faster, the saved file is small, and the quality is often close to full fine-tuning.

Say in interview: LoRA freezes the main model and trains only small added layers. It is cheap and fast, and the quality is close to full fine-tuning.

Q17. What is quantization?

Answer:

Quantization means storing the model numbers with fewer bits, for example 4-bit or 8-bit instead of 16-bit. The model becomes smaller and faster and uses less memory, so we can run it on cheaper hardware. The cost is a small drop in quality.

Say in interview: Quantization stores the model with fewer bits to make it smaller and faster, with a small loss in quality.

Q18. What is RAG and when do we use it instead of fine-tuning?

Answer:

RAG means we search our documents and give the related parts to the LLM in the prompt. We use RAG when the data is private, changes often, or when we want to show the sources. Fine-tuning changes the model itself and is used for style and behavior. Many projects use both.

Say in interview: RAG gives the model our data at question time without changing it. Fine-tuning changes the model for style. For facts I start with RAG.

Part 4: Behavior and Problems

Q19. What is hallucination? How do you reduce it?

Answer:

Hallucination means the model gives a wrong or made-up answer, but sounds confident. It happens because the model predicts likely words, it does not check facts.

- Use RAG, so the answer comes from real documents.
- Tell the model to answer only from the given context and to say 'I don't know' if not found.
- Ask for sources and verify them.
- Keep temperature low for factual tasks.
- Add checks and human review for important answers.

Say in interview: Hallucination is a confident wrong answer. I reduce it with RAG, clear prompts, low temperature, and source checking.

Q20. What are temperature and top-p?

Answer:

They control how random the output is. Low temperature gives predictable and consistent answers, good for facts and code. High temperature gives more creative and different answers. Top-p limits the choice to the most likely words that add up to probability p. We usually change one of them, not both.

Say in interview: Temperature and top-p control randomness. Low is for facts, high is for creative tasks.

Q21. How do you evaluate an LLM or a Gen AI application?

Answer:

There is no single number, so we use a mix:

- A test set of real questions with expected answers.
- Automatic checks, like comparing with the correct answer, or checking the format.
- LLM-as-a-judge: another LLM scores the answers. It is fast, but we must check it against human scores.
- Human review for quality.
- Metrics like accuracy, faithfulness (does it stick to the source), delay, and cost.
- Real user feedback after launch.

Say in interview: I use a test set, automatic checks, LLM-as-judge checked against humans, and real user feedback.

Q22. What are the risks and ethical issues of Gen AI?

Answer:

The main ones are:

- Hallucination (wrong information).

- Bias from the training data.
- Privacy and data leaks.
- Prompt injection and misuse.
- Copyright questions about training data and outputs.
- Fake content (deepfakes) and misinformation.

Say in interview: The main risks are wrong answers, bias, privacy, security attacks, and misuse. We handle them with guardrails, testing, and human review.

Q23. What are guardrails?

Answer:

Guardrails are checks around the model to keep it safe and on the topic. They can filter harmful or private content in the input and the output, block off-topic requests, check the format, and limit what tools the model can use. We keep them outside the model, in our code, so they are not easy to bypass.

Say in interview: Guardrails are safety checks on the input and the output of the model, to keep it safe, correct, and on the topic.

Part 5: Applications

Q24. What is an AI agent? What is MCP?

Answer:

An AI agent is an LLM that can use tools and take actions, like read or edit a database, search the web, or call an API. It works in a loop: think, call a tool, look at the result, and repeat until the task is done. A normal chatbot only answers once from its own knowledge.

- **MCP (Model Context Protocol)** is an open standard to connect AI apps with tools and data. We build one MCP server for a tool, and any MCP-supported AI app can use it, so we do not write a separate integration for every app.
- For risky actions like edit or delete, we ask the human to confirm first.

Say in interview: An agent is an LLM that uses tools in a loop to finish a task. MCP is a common standard to connect those tools to AI apps.

Q25. What are multimodal models, and how do image generation models work?

Answer:

Multimodal models can work with more than one type of data, like text, images, audio, and video, for example reading a photo and answering a question about it.

- **Diffusion models** (Stable Diffusion and others) create images by starting with random noise and removing the noise step by step until a clear image appears, guided by the text prompt.
- **GANs** use two networks: a generator that makes fake images, and a discriminator that tries to catch the fakes. They improve each other. Diffusion models are more popular now.

Say in interview: Multimodal models handle text, images, and audio together. Diffusion models create an image by removing noise step by step.

If you do not know an answer, say honestly: 'I have not worked on this directly, but my understanding is...' and then explain what you know. It is better than guessing.